

Structure, Not Texture: Train-Free, World-Knowledge-Grounded Interiors for Gaussian Splatting

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Abstract

When a 3D Gaussian-Splatting object is cut, broken, or sliced open, what does its interior look like? A splat is a hollow shell of surface primitives, so every existing answer *fabricates* the inside. PhysGaussian and PhysTalk deform only the surface and never expose an interior; FruitNinja hallucinates interior *texture* with a per-object diffusion optimization; GaussianFluent densifies interior gaussians with a diffusion prior in the loop. All of these priors are *learned or optimized*, blind to the identity of the specific object, and slow. We observe that a vision-language model already *knows* what is inside a watermelon, a tree log, or an onion, and that this knowledge is a *train-free, structural* prior. We present INSIDEOUT, a framework in which a VLM agent recognizes an object, reasons about its internal layout, and *writes a short program* that instantiates that structure directly in the gaussians—radial seeds, concentric rings, nested layers—which a topology-changing operation then reveals. There is no training, no per-object optimization, and no diffusion in the loop: interiors are authored in seconds and stay consistent under arbitrary cut directions. Because it is instant, the method scales editable, physically-changeable interiors from single objects to whole scenes—a step toward explorable worlds whose objects can be opened, cut, and broken. We demonstrate structurally-correct interiors across [TODO: N] objects and [TODO: M] operations, head-to-head against FruitNinja (structure vs. texture, zero-optimization runtime), GaussianFluent (no diffusion-in-loop), and PhysTalk (interior vs. none). We are explicit about the failure mode we inherit: when an object’s interior is genuinely uniden-

tifiable, the agent should *abstain* rather than confabulate.

1 Introduction

(Arc: (1) worlds from media should be changeable, not just viewable; (2) changing an object means exposing its inside; (3) splats are hollow shells, so the inside is always invented; (4) prior inventions are learned/optimized/slow/category-blind; (5) the VLM already knows the inside — use it as a train-free structural prior; (6) contributions.)

[TODO: one motivating figure: same watermelon splat cut open — PhysTalk (deflates, no interior) | FruitNinja-style texture fill (smeared, wrong seed placement, cut-inconsistent) | ours (radial seeds placed by world knowledge, consistent under any cut).]

Contributions.

- We reframe an object’s interior as recoverable from a VLM’s *world knowledge*, used as a *train-free structural prior* — in place of the learned, optimized, or diffusion priors of prior work.
- An *agentic, train-free* pipeline: recognize → reason about internal layout → author a program over an interior/physics op-library → instantiate structured interior gaussians → reveal via a topology-changing operation. No per-object optimization, no diffusion in the loop.
- *Structure, not texture*: explicit 3D internal geometry (seeds, rings, layers) that is *consistent under any cut direction by construction*, unlike per-cross-section inpainting.

- A demonstration across objects and operations with direct head-to-heads against FruitNinja, GaussianFluent, and PhysTalk, and an honest treatment of *unidentifiable* interiors (abstention), connecting to identifiability in single-view 3D generation.

2 Related Work

(Position surgically — the gap is narrow; each neighbor is one axis away. State plainly what each does and does NOT do.)

Physics on Gaussian Splats. PhysGaussian, PhysSplat, and **PhysTalk** embed 3DGS in a physics simulator (MPM / Genesis) and deform the visible surface. PhysTalk is the closest on the *agentic, train-free* axis—an LLM writes physics code—but it never instantiates or reveals an interior; it transfers particle motion back to existing surface gaussians. [TODO: cite; confirm no topology-change / interior].

Interior generation for splats. FruitNinja (CVPR’25) first revealed a splat interior under slicing, via *per-object* progressive 2D-diffusion inpainting on cross-sections plus voxel smoothing. It is not agentic, its prior is category-blind, and its result is *texture* with no guarantee that seeds/rings sit where they should, nor that different cuts agree. **GaussianFluent** (ICLR’26) densifies interior gaussians for brittle *fracture*, but with a *diffusion prior in the loop*, not agentic and not identity-grounded. [TODO: cite both; this is the head-to-head axis.]

Inferring interior structure. TopoGaussian infers an object’s *load-bearing* interior topology by optimizing a representation so a simulator explains observed motion—per-object optimization, not semantic (“what mechanically works”, not “this is a watermelon”). [TODO: cite.]

Agentic program authoring for 3D. Articraft (2026) has an LLM agent write a program against a DSL to build articulated 3D *meshes*. It establishes

“agent-writes-a-program-for-3D” as a paradigm—so that framing alone is not our novelty—but it does not touch captured splats, interiors, or topology-changing reveals. [TODO: cite; use to pre-empt “just Articraft for splats”.]

Editable / language-driven splats. [TODO: REALM (MLLM-agent surface editing of 3DGS, no interior), VR-GS (tet-proxy interactive physics), etc.]

3 Method

(Reuse the physstalk substrate: load/render, solidify (axis-stab interior fill), op-library (shatter/slice/break), agent-authoring core. The NEW part is the structured interior authored from a VLM schema.)

3.1 Overview

Given a captured 3DGS object and a natural-language instruction (“cut it open”, “smash it”), a VLM agent produces (i) an *interior schema* describing the object’s internal structure and (ii) a *transform program* over an op-library. The program instantiates the structured interior as gaussians and applies a topology-changing operation that reveals it. [TODO: pipeline figure.]

3.2 World knowledge as a structural prior

The agent recognizes the object and emits a compact *interior schema*—a small typed program, e.g. `radial_seeds(count, size, region), concentric_rings(spacing, colors), layers(thicknesses, colors), solid_core(color), hollow(shell_thickness)`. (The schema is the train-free prior: it encodes what the VLM knows, and is category-correct by construction rather than by inpainting luck.) [TODO: define the schema grammar / op-library table.]

3.3 Instantiating structured interior gaussians

We fill the hollow shell with interior gaussians (axis-stabbing occupancy test, cf. §3) and *place structure* according to the schema—seeds as discrete clusters at radial positions, rings as concentric shells, layers as nested strata—rather than copying surface color inward. [TODO: algorithm box; contrast with FruitNinja’s OpaqueAtom texture fill.]

3.4 Topology-changing operations

[TODO: slice / fracture / break / bite / drill from the op-library; the reveal; two-sided consistency for fracture.]

4 Experiments

(Framed as a CAPABILITY DEMONSTRATION, not a benchmark. NO ground-truth interior exists (FruitNinja says so) — we do NOT chase faithfulness-to-truth. Success = structure beats texture, train-free.)

4.1 Success criteria

- **Structural correctness** (primary): do placed elements match category-expected layout (seeds radial, rings concentric, layers nested)? Blind human + VLM-judge (randomized order, report both directions — VLM judges have position bias), plus a geometric check (discrete 3D seeds at plausible positions vs. smeared texture).
- **Cut-consistency**: same object, multiple cuts → interior stays 3D-consistent (no GT needed).
- **Zero-optimization runtime**: seconds vs. minutes — the one unarguable number.

4.2 Dataset

[TODO: ~30–40 objects via TripoSplat / image→3DGS, weighted to visibly-structured interiors: watermelon (radial seeds), tree log (rings), onion/cabbage (layers), pomegranate (arils), orange (segments), cake (strata); a few real captured objects; plus deliberately *ambiguous*

interiors (sealed gadget, random rock) for the abstention/failure section. Category-reference cut photos for judging — NOT per-instance GT.]

4.3 Head-to-heads

[TODO: vs FruitNinja (structure vs. texture + runtime); vs GaussianFluent (no diffusion-in-loop, fracture); vs PhysTalk (interior vs. none). Skipping any → reviewer reduces us to that method.]

5 Limitations

When an object’s interior is genuinely *unidentifiable* from its exterior (a sealed device, an arbitrary rock), no prior—VLM or otherwise—can recover it; the agent should *abstain* rather than confabulate. (This is where the unidentifiability thesis of our prior work reappears — as an honest limitation, not the paper’s thrust.) Our interiors are *plausible and structurally category-correct*, not faithful to the specific instance; and the VLM’s knowledge has gaps for rare or novel objects. [TODO: expand.]

6 Conclusion

[TODO: One instant, train-free agent authors structured interiors from world knowledge — the enabling piece for scaling changeable objects from one to a whole explorable world.]